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Investment Intent Toward Sustainable Waste-to-Energy Deployment: Survey-Based Logistic Regression Evidence from India

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Abstract: Municipal solid waste management remains a critical sustainability challenge in India, while waste-to-energy (WtE) systems increasingly attract attention as a pathway to simultaneously support waste reduction, resource recovery, and sustainable energy development. However, investment participation in WtE projects remains inconsistent despite growing policy interest and deployment targets. This study investigates the factors associated with investment intent toward WtE projects and examines how project perceptions influence early-stage investment attractiveness in the Indian context. A structured questionnaire was administered to 123 respondents, including students and early-career professionals from energy-related and service-oriented sectors. Seven dimensions of project perception—financial, informational, regulatory, market, environmental, socio-cultural, and technological—were evaluated. Fisher's Exact Test and binary logistic regression were employed to examine factor associations and estimate their relative influence on investment intent. The results showed that all seven factors were significantly associated with investment intent at the 5% significance level. The logistic regression model achieved an overall prediction accuracy of 83.67% on the holdout sample, with a precision of 81.08%, recall of 96.77%, and an F1 score of 0.882 for the investment-positive class. Market acceptance and environmental credibility emerged as the strongest positive predictors of investment willingness, whereas information barriers and perceptions of high capital intensity reduced investment likelihood. The findings indicate that investment decisions related to WtE deployment extend beyond expected financial returns and are strongly influenced by perceptions of market feasibility, environmental reliability, and project transparency. This study demonstrates that improving information quality, strengthening market confidence, and reinforcing environmental assurance can enhance the investment readiness and bankability of sustainable WtE systems. The study contributes empirical evidence for supporting sustainable energy deployment strategies and improving project evaluation practices in emerging economies.

Keywords: Waste-to-energy; Sustainable energy investment; Investment intent; Energy sustainability; Project bankability; Logistic regression; India

1 Introduction

Municipal solid waste management in India continues to be dominated by collection and disposal practices that were not designed for the scale, composition, or complexity of present-day urban waste streams. Rapid urbanisation, rising consumption, and weak source segregation have produced a situation in which a significant share of waste is collected in larger cities, but a much smaller share is processed through environmentally sound systems [1]. Large volumes of mixed waste are still directed to uncontrolled or weakly managed dump sites, creating persistent risks for land, water, air quality, and public health.

Waste-to-energy (WtE) technology has consequently attracted increasing attention from policymakers and investors because it offers the possibility of addressing waste disposal and energy recovery simultaneously. Depending on the feedstock and conversion pathway, WtE projects may produce electricity, compressed biogas (CBG), or refuse-derived fuel while also reducing landfill dependence [2]. Although this combination gives WtE a compelling role in circular economy strategies, actual project performance and investment outcomes remain highly uneven.

The Indian context illustrates this gap clearly. National policy initiatives have acknowledged the strategic relevance of bioenergy and alternative fuels, and ambitious deployment targets have been announced for CBG and associated

waste-processing infrastructure [3]. However, realised project capacity remains considerably below stated policy ambition. The resulting gap between technical potential and committed capital points to a broader investment challenge rather than a purely technological one.

Investment decisions in WtE projects are shaped by a wide range of concerns, including project cost, technology maturity, policy stability, environmental compliance, social acceptance, and the reliability of offtake for the energy or by-products produced [4, 5]. Several Indian WtE projects have also underperformed because plant designs and business assumptions were not adequately aligned with local waste characteristics [5, 6]. As a result, investment intent is influenced by both quantifiable economic considerations and broader perceptions of risk and credibility.

The present study examines how these perceptions relate to willingness to invest in WtE projects. Using survey evidence and statistical modelling, it identifies the factors most strongly associated with positive investment intent and translates those findings into practical guidance for improving project bankability in the Indian market.

2 Literature Review

Published work on WtE investment barriers tends to converge around a common set of themes even when studies examine different countries or technologies. Technical feasibility, economic viability, regulatory support, environmental performance, and social acceptance appear repeatedly across the literature as the most consequential determinants of project success [1, 2, 7].

2.1 Financial and Economic Factors

High capital intensity is frequently identified as a primary barrier to WtE investment. Compared with conventional waste disposal methods, WtE projects require substantial upfront capital, making project viability highly sensitive to financing structure and perceived risk [1, 4]. In India, long payback periods, cost overruns, and uncertainty in revenue realisation have discouraged private sector involvement [4, 5]. Evidence from agricultural biogas projects in Europe similarly highlights that financial feasibility and perceived downside risk strongly shape early-stage investment intent [8].

2.2 Information and Knowledge Barriers

Information asymmetry is recognised as a critical non-financial barrier affecting WtE project attractiveness. Investors may support the sustainability rationale of WtE while remaining uncertain about technology performance, feedstock quality, operating risks, and regulatory compliance [4, 9]. Poor dissemination of project-specific information has been shown to reduce confidence even when supportive policies are in place [7, 10]. Ravindranath and Balachandra [9] emphasise that fragmented or incomplete information increases investor conservatism in bioenergy projects.

2.3 Regulatory and Institutional Framework

Regulatory uncertainty significantly influences perceived WtE project risk. In India, institutional fragmentation among municipal bodies, state regulators, and central authorities complicates waste supply contracts and tariff determination [4, 11]. Misalignment between national renewable energy objectives and state-level tariff implementation creates uncertainty at the project appraisal stage [4]. Studies of public-private partnership WtE projects in China further show that regulatory inconsistency and government decision-making risk materially affect project performance and investor confidence [12].

2.4 Market Acceptance and Revenue Security

Market conditions play a central role in WtE investment decisions. Electricity or CBG produced from waste often competes with lower-cost fossil fuels, particularly in the absence of long-term offtake agreements or mandated demand [4, 13]. Without secure revenue pathways, technically viable projects may struggle to attract capital [6, 13]. Empirical studies demonstrate that market assurance is a key determinant of perceived project bankability in developing economies [7].

2.5 Environmental Performance and Risk Perception

Although WtE technologies offer environmental benefits such as landfill diversion and emissions reduction, perceived environmental risks strongly influence investment intent. Concerns related to emissions, odour, ash handling, and groundwater contamination frequently lead to public opposition and stricter regulatory scrutiny [5, 12]. Empirical studies on waste treatment acceptance show that environmental credibility is closely linked to long-term project legitimacy and operability [14]. Past failures of WtE plants in India have heightened sensitivity toward environmental safeguards, reinforcing investor caution [5].

2.6 Socio-Cultural Acceptance

Social acceptance complements technical and regulatory considerations in shaping WtE project outcomes. Public resistance, often associated with the "Not In My Back Yard" (NIMBY) phenomenon, has affected WtE plant siting and operation in multiple countries [12, 14]. Weak community engagement increases reputational and political risk, indirectly affecting investment attractiveness [5].

2.7 Technological Factors

Technology choice and perceived reliability influence WtE investment intent, particularly in developing markets. Concerns include feedstock variability, dependence on imported equipment, limited access to skilled technical support, and lack of proven large-scale operation [4, 6]. Studies indicate that poor matching of technology to local waste characteristics has contributed to underperformance of several Indian WtE plants [5].

While prior literature identifies multiple barriers to WtE deployment, empirical studies quantifying how these factors jointly shape investment intent remain limited. This study addresses this gap by applying statistical analysis to perception-based data to support early-stage investment screening and managerial decision-making.

3 Conceptual Framework, Objectives and Hypotheses

3.1 Conceptual Framework

The conceptual model used in this study treats investment intent as the dependent variable and seven barrier categories as independent variables: financial, informational, regulatory, market, environmental, socio-cultural, and technological. Each factor represents a set of perceptions that can either improve or weaken the attractiveness of a proposed WtE project.

The logic of the model is straightforward. A respondent is more likely to support investment when the project appears commercially viable, environmentally defensible, understandable, and institutionally supported. A respondent is less likely to support investment when the project appears technically uncertain, socially contested, or poorly explained. The seven hypotheses tested in this paper reflect that structure.

Figure 1 presents the conceptual framework linking the seven barrier categories to the binary investment decision outcome.

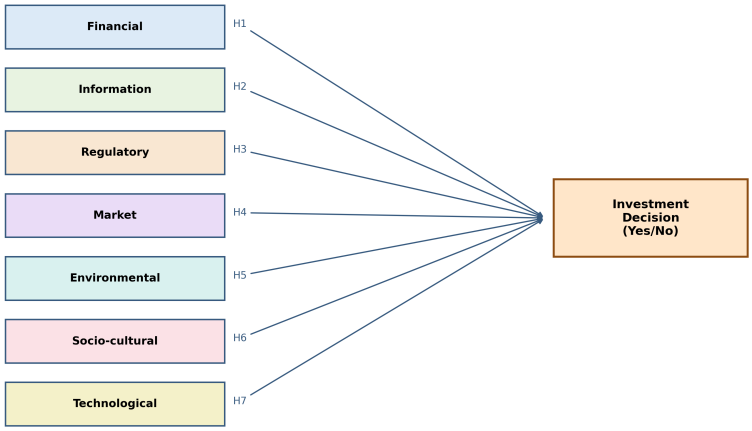


Figure 1. Conceptual framework linking seven barrier categories to investment decision

Note: The factor structure is based on previous studies [1, 4, 7].

3.2 Research Objectives

1. To identify the factors associated with investor willingness to fund WtE projects in India using Fisher's Exact Test and logistic regression.

2. To translate the empirical findings into practical guidance for project developers seeking to improve investment readiness.

3.3 Hypothesis

H1 (Financial factor): There is an association between investment intent and the financial barrier factor.

H2 (Information factor): There is an association between investment intent and the information barrier factor.

H3 (Regulatory factor): There is an association between investment intent and the regulatory framework factor.

H4 (Market factor): There is an association between investment intent and market acceptance of WtE outputs.

H5 (Environmental factor): There is an association between investment intent and perceived environmental performance of WtE projects.

H6 (Socio-cultural factor): There is an association between investment intent and social acceptance of WtE projects.

H7 (Technological factor): There is an association between investment intent and the technology deployment factor.

The factor categories and hypotheses presented here are developed from and consistent with the author's prior dissertation research on WtE investment factors in India.

4 Research Methodology

A cross-sectional survey design was adopted for the empirical part of the study. Respondents were presented with a scenario describing a proposed WtE plant requiring approximately INR 25 crore in total funding, with a projected return of 21%, a six-year payback period, and regulatory approvals stated to be in place. They were then asked whether they would consider investing in the project and to rate a series of statements linked to the seven barrier categories on a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

The questionnaire was pilot-tested with eight respondents (four students and four working professionals) prior to full data collection. Two items were revised for clarity based on pilot feedback before the final survey was administered. Survey items were adapted from established WtE and bioenergy barrier literature [4–7, 10] and contextualised for the Indian WtE setting. Table 1 provides a complete listing of all survey items by factor category.

Table 1. Survey instrument items by factor category

Factor	Item No.	Survey Statement	Scale
Financial	Q1	Such projects are highly capital intensive.	1–5 Likert
Financial	Q2	High up-front installation costs are a barrier to investment.	1–5 Likert
Information	Q3	Lack of proper knowledge or information is a barrier.	1–5 Likert
Regulatory	Q4	A proper regulatory framework supports such projects.	1–5 Likert
Regulatory	Q5	Government intervention affects such projects.	1–5 Likert
Regulatory	Q6	Non-alignment of national and state tariff policies is a barrier.	1–5 Likert
Market	Q7	Compressed biogas (CBG) can be treated as a viable alternate fuel.	1–5 Likert
Environmental	Q8	Projects have an environmental impact if not operated properly.	1–5 Likert
Environmental	Q9	Projects have a considerable negative impact on the environment.	1–5 Likert
Socio-cultural	Q10	There is a lack of social acceptance for biogas from waste.	1–5 Likert
Socio-cultural	Q11	There is a lack of acceptance of waste-to-energy (WtE) technology by society.	1–5 Likert
Technological	Q12	Poor dissemination of technology information is a barrier.	1–5 Likert
Technological	Q13	There is a lack of access to appropriate technology.	1–5 Likert
Technological	Q14	There is a lack of proven large-scale technology.	1–5 Likert

Note: The items were adapted from sources [4–7, 10] and contextualised for the Indian WtE setting. Scale: 1 = Strongly Disagree, 5 = Strongly Agree.

The final dataset consisted of 123 valid responses collected through a convenience sampling approach via an online survey platform. Of the 123 respondents, 31 (approximately 25%) were students enrolled in energy, engineering, or management programmes, and 92 (approximately 75%) were early-career professionals with one to five years of experience across the energy, IT, consulting, and marketing sectors. Age distribution was concentrated between 22 and 28 years, with 25 years representing the single most common age in the sample. This respondent composition was deliberate: the study targets perception-based investment intent at the early project screening stage, where assessments by informed but non-institutional respondents provide useful directional evidence about which project attributes are likely to be prioritised by early-stage evaluators. It should be noted that this sample does not represent the full population of Indian institutional investors or infrastructure funds; findings should therefore be interpreted as indicative rather than definitive, and replication with larger and more diverse investor samples is recommended.

Because the dependent variable was binary, and several cross-tabulation cells contained small counts, Fisher's Exact Test [15] was used to test associations between each factor and investment intent. A p-value of 0.05 or below was treated as statistically significant. Binary logistic regression [16] was then used to estimate the direction and relative strength of each factor in predicting investment intent. The model was trained on 60% of the observations (74 instances) and evaluated on a 40% holdout set (49 instances).

5 Results and Analysis

The respondent profile shows a sample weighted toward professionals (75%) over students (25%), with age distribution concentrated in the mid-twenties. Figures 2–3 present the respondent background and age distribution respectively. This profile is useful for interpreting the results because it suggests that most responses came from individuals who are early in their careers but already accustomed to analysing projects or business cases.

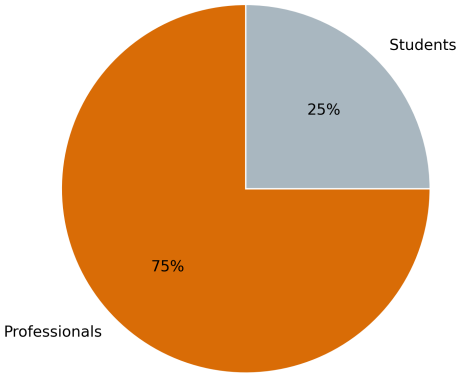


Figure 2. Respondent profile by background
Note: Data from author's survey (n = 123).

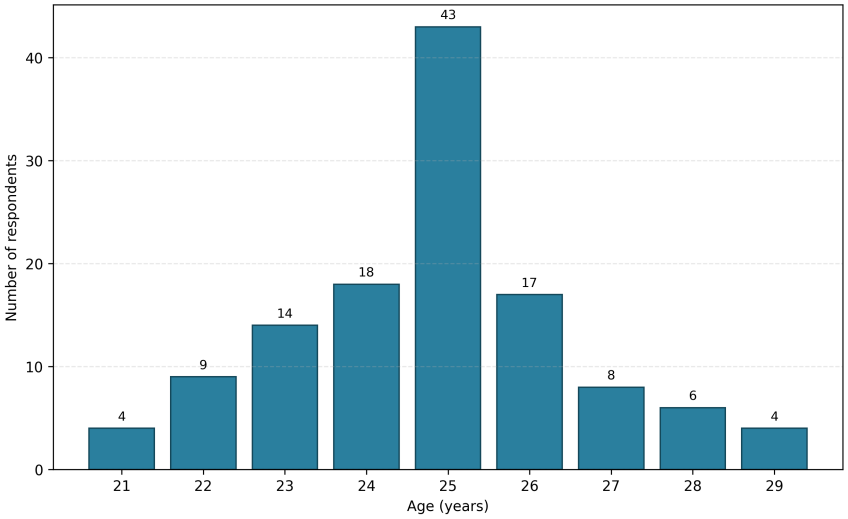


Figure 3. Age distribution of respondents
Note: Data from author's survey (n = 123).

Among the eight project risk categories presented to respondents, technical and operational risk ranked highest at 76%, followed by project selection and finance risk (63%), and political and legal risk (56%). Figure 4 presents the full risk factor ranking. This result is consistent with the Indian WtE context, where investors remain cautious about feedstock quality, plant uptime, process reliability, and technology suitability. Respondents do not view WtE as purely a financing problem but as a project execution challenge with multiple interconnected uncertainties [5].

The perception data also shows strong agreement that WtE projects are capital intensive and that lack of proper knowledge is a major barrier. Respondents were more mixed on questions of market viability and social acceptance, suggesting that these areas remain contested and may depend heavily on project location, technology type, and the credibility of specific implementation plans.

The perception data in Table 2 carry important practical implications for project developers. Strong agreement that WtE projects are capital intensive (Q1, Q2) confirms that high upfront cost is a well-established baseline expectation among respondents; developers therefore cannot reduce hesitancy simply by highlighting financial returns alone. The equally strong agreement that lack of information is a barrier (Q3) points to a more actionable gap: opacity in project

documentation acts as a structural deterrent that can be addressed through transparent, technically grounded investor materials. Mixed responses on market viability (Q7) and social acceptance (Q10, Q11) suggest that these dimensions are most sensitive to the specific project context, highlighting the importance of site-specific market assessments and community engagement strategies.

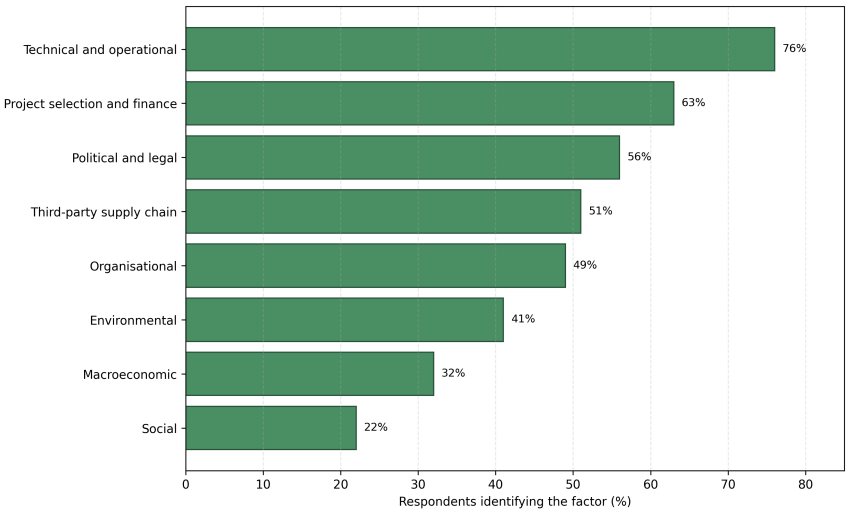


Figure 4. Risk factors in waste-to-energy (WtE) projects identified by respondents

Note: Data from author's survey (n = 123).

Table 2. Respondent perception of barrier factors in waste-to-energy (WtE) projects

Factor	Statement	Frequency (Modal)	Interpretation
Finance	Such projects are highly capital intensive	48 (rating: 5)	Strongly agree
Finance	High up-front installation costs are a barrier	50 (rating: 5)	Strongly agree
Information	Proper knowledge or information is a barrier	60 (rating: 5)	Strongly agree
Regulatory	A proper regulatory framework supports such projects	48 (rating: 4)	Agree
Regulatory	Government intervention affects such projects	40 (rating: 3)	Undecided
Regulatory	Non-alignment of national and state tariff policies is a barrier	40 (rating: 4)	Agree
Market	CBG can be treated as a viable alternate fuel	38 (rating: 3)	Undecided
Environmental	Projects have environmental impact if not operated properly	50 (rating: 5)	Strongly agree
Environmental	Projects have a considerable negative impact on the environment	47 (rating: 2)	Disagree
Socio-cultural	Lack of social acceptance for biogas from waste	40 (rating: 3)	Undecided
Socio-cultural	Lack of acceptance of WtE technology by society	37 (rating: 3)	Undecided
Technological	Poor dissemination of technology information is a barrier	40 (rating: 4)	Agree
Technological	Lack of access to appropriate technology	45 (rating: 5)	Strongly agree
Technological	Lack of proven large-scale technology	38 (rating: 3)	Undecided

Note: Frequency indicates the number of respondents selecting the modal response value shown. Rating scale: 1 = Strongly Disagree, 5 = Strongly Agree.

Fisher's Exact Test rejected the null hypothesis in all seven cases, indicating statistically significant associations between each factor and investment intent (Table 3). The strongest associations were observed for regulatory (χ^2 = 68.223), environmental (χ^2 = 62.802), market (χ^2 = 52.408), and technological (χ^2 = 52.209) factors. These results justify the move to multivariate modelling and reinforce the idea that investment intent is shaped by a combination of institutional, technical, and perceptual variables. Notably, even the financial factor (H1, Fisher sig. = 0.022) and information barrier (H2, Fisher sig. = 0.001) showed significant associations, though with comparatively weaker χ^2 values, suggesting they play a supportive rather than dominant role in investment decision-making.

The fitted logistic model can be expressed as:

$$P(\text{investment} = 1) = \frac{1}{1 + \exp\left(\begin{array}{l} 6.269 - 1.075 \cdot \text{Market} - 0.901 \cdot \text{Environmental} + 0.614 \cdot \text{Information} \\ - 0.432 \cdot \text{Regulatory} - 0.597 \cdot \text{Social} - 0.119 \cdot \text{Tech} + 0.115 \cdot \text{Capital} \end{array}\right)}$$

The logistic regression results (Table 4) carry direct implications for project developers and policymakers. Market

Table 3. Fisher's Exact Test results for factor association with investment intent

Hypothesis	Factor	Fisher sig.	χ^2	Decision
H1	Financial	0.022	11.334	Reject H_0
H2	Information	0.001	16.339	Reject H_0
H3	Regulatory	0.000	68.223	Reject H_0
H4	Market	0.000	52.408	Reject H_0
H5	Environmental	0.000	62.802	Reject H_0
H6	Socio-cultural	0.000	47.537	Reject H_0
H7	Technological	0.000	52.209	Reject H_0

Note: All tests two-sided; df = 4 for each test. Significance threshold $p \leq 0.05$.

acceptance (B = 1.075, Exp(B) = 2.931) is the strongest positive predictor: each unit increase in perceived market viability nearly triples the odds of investment intent, underscoring that revenue security through long-term offtake agreements and mandated CBG demand is central to project bankability. Environmental credibility (B = 0.901, Exp(B) = 2.462) is nearly as influential, suggesting that investors treat perceived environmental performance not as a secondary reputational issue but as a proxy for regulatory continuity and operational risk—projects that cannot demonstrate credible environmental safeguards are assessed as structurally more exposed to compliance risk and public opposition. Information barriers (B = -0.614, Exp(B) = 0.541) reduce the probability of investment even when the financial case appears sound, implying that opacity in project documentation acts as a standalone deterrent independent of underlying project quality. The comparatively weaker coefficient on capital intensity (B = -0.115) is not evidence that cost is unimportant; rather, respondents appear to treat high capital expenditure as a normal characteristic of WtE infrastructure, and the financing burden becomes tolerable when supported by a credible market and environmental story.

Table 4. Logistic regression coefficients

Variable	B	S.E.	Wald	Exp(B)
Highly capital intensive	-0.115	0.417	0.076 ($p = 0.782$)	0.891
Information barrier	-0.614	0.379	2.625 ($p = 0.105$)	0.541
Regulatory framework	0.432	0.397	1.186 ($p = 0.276$)	1.541
Market acceptance	1.075	0.445	5.837 ($p = 0.016$)	2.931
Environmental friendly	0.901	0.353	6.514 ($p = 0.011$)	2.462
Socially accepted	0.597	0.437	1.866 ($p = 0.172$)	1.817
Technological knowledge	0.119	0.347	0.117 ($p = 0.732$)	1.126
Constant	-6.269	1.709	13.453 ($p = 0.000$)	0.002

Note: Dependent variable = investment intent (Yes = 1, No = 0). Model estimated on training set (n = 74; 60%) evaluated on holdout set (n = 49; 40%). Nagelkerke R^2 = 0.756.

The holdout sample contains 31 investment-positive and 18 investment-negative instances, reflecting a moderate class imbalance (63%/37%) (Table 5). The higher recall for the positive class indicates the model is conservative—it rarely misses a true investor—but this comes at some cost to specificity, as evidenced by 7 false positives. The overall accuracy of 83.67% should therefore be read alongside the class-level precision and recall metrics above.

Table 5. Confusion matrix for the holdout sample (n = 49)

	Predicted: Yes	Predicted: No	Class Recall
Actual: Yes	30	1	96.77%
Actual: No	7	11	61.11%
Class Precision	81.08%	91.67%	83.67% (Acc.)

Note: Precision (investment-positive class) = 30/37 = 81.08%; Recall (investment-positive class) = 30/31 = 96.77%; F1 score (investment-positive class) = 0.882. Precision (investment-negative class) = 11/12 = 91.67%; Recall (investment-negative class) = 11/18 = 61.11%; F1 score (investment-negative class) = 0.733.

6 Discussion

The results indicate that investors do not respond to WtE proposals on the basis of sustainability claims alone. They respond more strongly to evidence that a project can operate reliably, place its outputs in a credible market, and

satisfy environmental scrutiny. This helps explain why market acceptance emerged as the strongest positive predictor. When the offtake pathway remains uncertain, even technically promising projects are less likely to appear investable.

Environmental credibility is nearly as influential, which suggests that environmental performance is not viewed as a secondary reputational issue but as part of the project's fundamental risk profile. Concerns relating to emissions, odour, ash handling, groundwater contamination, and long-term compliance shape expectations about operating continuity and reputational exposure. Respondents appear to distinguish clearly between projects that merely invoke sustainability language and those that can demonstrate operational safeguards.

The negative contribution of information barriers is similarly important. Many WtE projects fail to convert general interest into investment because the investment case is not explained with sufficient technical and commercial clarity [4, 10]. Investors require credible information on feedstock assumptions, process configuration, revenue logic, regulatory status, and contingency planning. When those elements are incomplete or difficult to verify, hesitation increases even in cases where the project concept is attractive.

The comparatively weaker direct coefficient on capital intensity should not be interpreted as evidence that cost is unimportant. A more plausible interpretation is that respondents already regard high capital expenditure as a normal characteristic of WtE infrastructure. Under those conditions, project cost becomes acceptable when it is supported by a convincing business case, appropriate risk allocation, and credible market demand [8].

For project developers, the practical implication is that investment readiness requires more than a favourable financial projection. It requires coordinated communication across technology, operations, compliance, and market positioning. For policymakers, the findings reinforce the value of stable tariff structures, dependable waste supply arrangements, and clearer support mechanisms for projects that align waste management objectives with clean energy and alternative fuel goals [9, 11].

7 Ethical Considerations

The study involved a voluntary, anonymous questionnaire survey without collection of sensitive personal data. All participants were informed of the study purpose and provided informed consent prior to responding. Given the minimal-risk nature of the research, formal institutional ethics approval was not required.

8 Conclusion

This study shows that investment intent in Indian WtE projects is shaped by a broad set of interacting concerns rather than a single dominant barrier. Financial, informational, regulatory, market, environmental, socio-cultural, and technological factors all showed statistically significant associations with willingness to invest.

Within the predictive model, market acceptance and environmental friendliness stood out as the strongest positive influences, while information barriers and perceptions of heavy capital intensity reduced investment likelihood. These findings suggest that developers who want to improve project bankability should focus first on demonstrating a reliable revenue pathway, building trust in environmental controls, and presenting the project in a more transparent and technically grounded way.

The study is limited by its convenience sample and by the fact that the data capture stated intent rather than actual investment transactions. Even so, the results provide a useful evidence base for improving how WtE projects are framed, evaluated, and financed in India. Future work can build on this by using larger samples drawn from institutional investors and infrastructure funds, comparing state-level policy environments, and testing whether the same determinants hold for actual investment transactions.

Limitations

The study relies on students and early-career professionals rather than institutional investors and reflects stated investment intent rather than realised investment behaviour. The sample of 123 respondents is dominated by students and early-career professionals aged 22–28, and was collected using a convenience sampling technique, which limits external validity. The respondents' backgrounds—primarily energy, IT, consulting, and marketing—provide useful early-stage screening perspectives but do not represent the decision-making processes of institutional investors, infrastructure funds, or bank credit committees. The findings are therefore exploratory and should not be generalised to the full population of Indian WtE investors. They are best suited to informing early-stage project communication strategies and identifying the most salient perception dimensions for further investigation with more representative samples.

Ethical Approval

The study involved voluntary survey participation with no collection of sensitive personal data, and informed consent was obtained from all participants.

Data Availability

The survey dataset can be shared by the corresponding author upon reasonable request.

Conflicts of Interest

The author declares no conflicts of interest.

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